TCS AI ENGINEERING HACKATHON 2026 · CHALLENGE 05

AstraRoute AI

Autonomous AI FinOps & Intelligent Model Routing

The cheapest model is not always right. The most powerful model is not always necessary.



COST
Unnecessary expensive inference

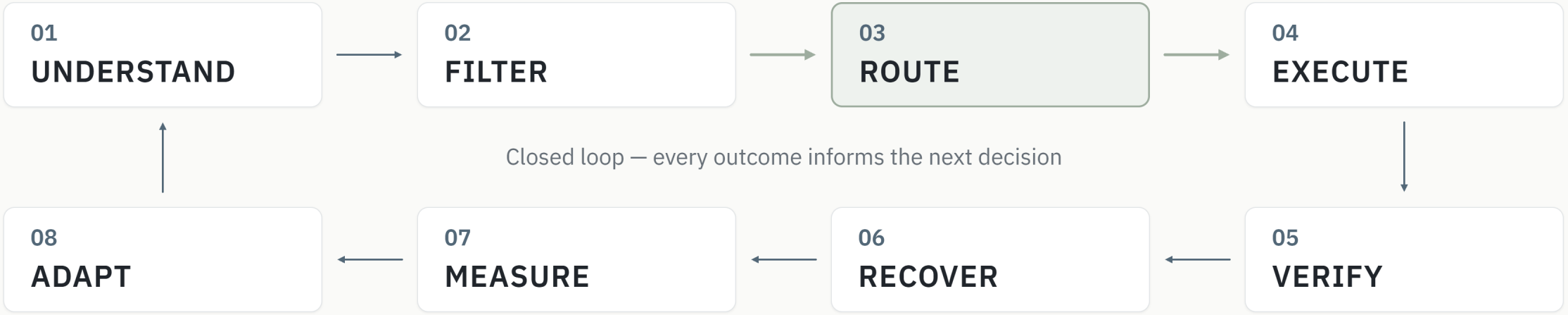
QUALITY
Cheap routing can reduce quality

LATENCY
Extra calls increase response time

GOVERNANCE
Privacy, budget and criticality matter

Team MindX

One Control Loop for Every AI Request



SLM
Fast pre-routing

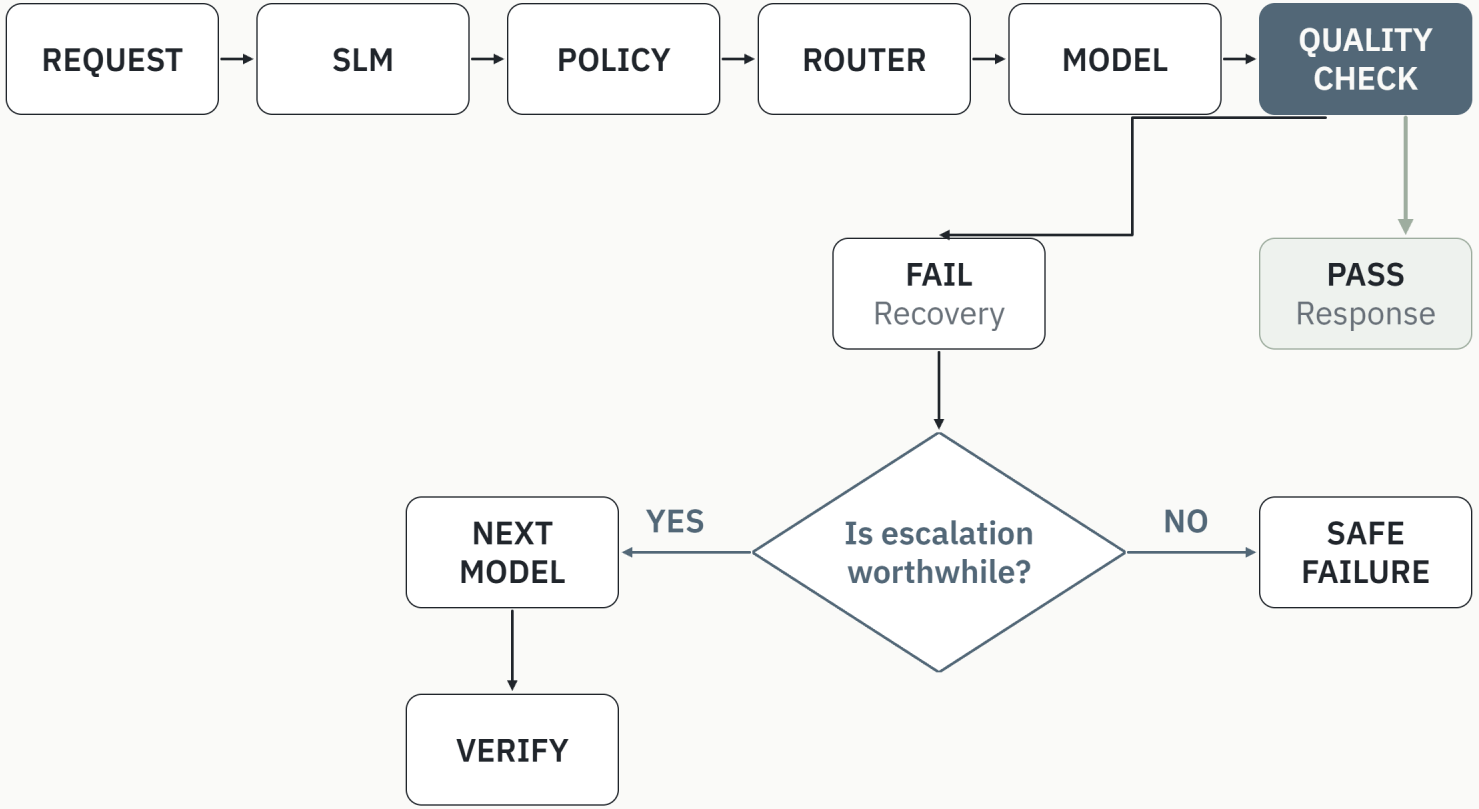
POLICY
Eligibility + constraints

ROUTER
Cost / quality / latency decision

GUARDIAN
Verify + recover

FEEDBACK FinOps telemetry → Historical performance → Future routing decisions

The Difference Happens When a Model Fails



Pass → respond · Fail → re-decide

EXAMPLE

Required quality

85%

Medium model **62% · FAIL**

Recovery checks

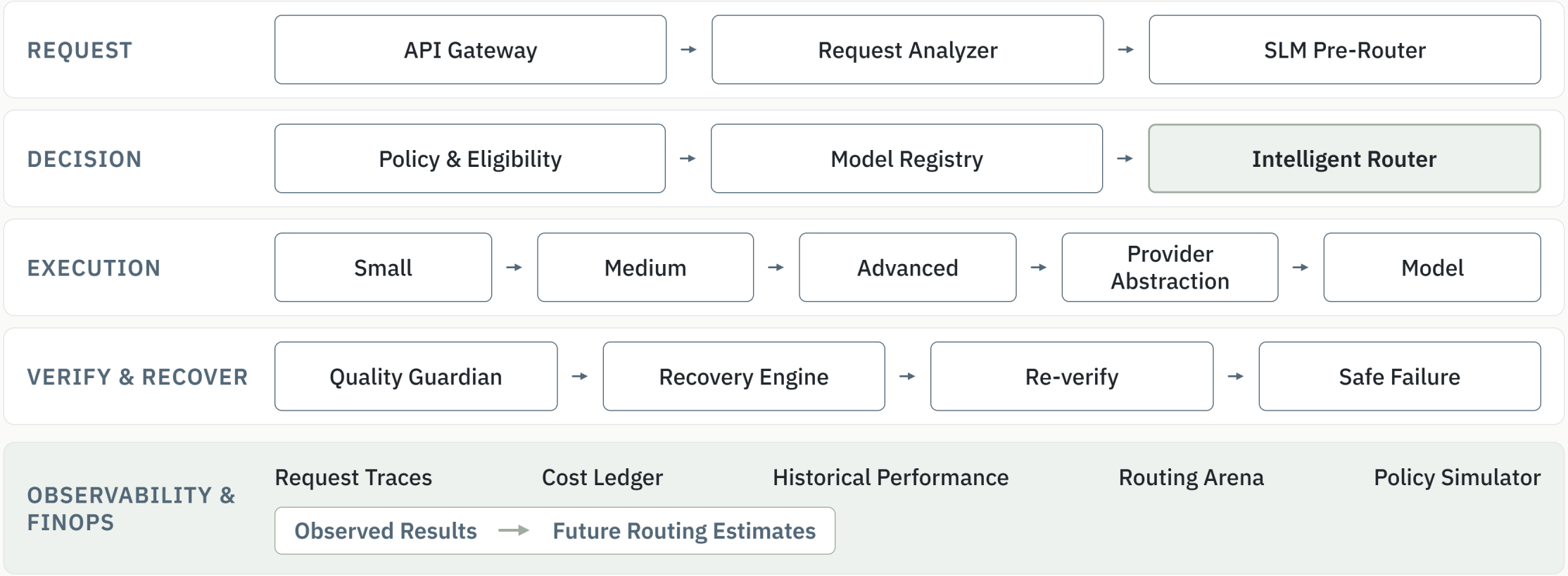
Eligible model · budget · latency · criticality · expected quality gain

Advanced model **93% · PASS**

Escalation is triggered only when expected improvement justifies additional cost and latency.

04 — SYSTEM ARCHITECTURE

Inside AstraRoute AI



TECH STACK React · Tailwind · FastAPI · Python · PostgreSQL/Supabase · SLM + LLM APIs · Pandas · Docker

From AI Cost Management to Closed-Loop AI Control

INDUSTRY

FinOps
Cost / budgets / optimization

AI Gateways
Routing / caching / governance

Observability
Usage / latency / reliability

Model Evaluation
Quality measurement

ASTRAROUTE — CONNECTS THEM

- **POLICY**
- **ROUTE**
- **EXECUTE**
- **VERIFY**
- **RECOVER**
- **MEASURE**
- **ADAPT**

Four differentiators

- **Quality-aware FinOps**
- **Economic recovery**
- **Cost per successful outcome**
- **Evidence-driven adaptation**

TCS ALIGNMENT

REASON	Understand request
PLAN	Select strategy
ACT	Execute
VERIFY	Validate outcome
ADAPT	Learn from telemetry

“We don’t claim to invent the individual capabilities. We connect them into a request-level closed loop where AI economics directly influence execution decisions.”

AstraRoute AI — Optimize the cost of achieving a verified AI outcome.